LUKE PAYNE

SOFTWARE ENGINEER | PYTHON, TYPESCRIPT AND APPLIED AI

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Software engineer building Python APIs and TypeScript/Node.js applications, with C++ systems and ML data-pipeline experience. Applied AI work includes retrieval, structured LLM tool use and automated evaluation.

TECHNICAL SKILLS

Languages and backend: Python, TypeScript, C++, SQL, Node.js, FastAPI, REST APIs
AI applications: LangChain, LangGraph, Ollama, retrieval/RAG, Chroma, tool calling
Engineering: Docker, Git, pytest, Playwright, CI/CD, data pipelines, authentication

PROFESSIONAL EXPERIENCE

ADNET Systems | Python Software Developer — Sep 2025 – Jun 2026

- Contributed to software development, testing, and observability for scientific data applications.

Verint | Software Engineer — Nov 2023 – Nov 2024

- Developed customer-facing B2B SaaS features in TypeScript and Node.js; investigated defects and implemented product enhancements.
- Worked with product and engineering teammates through code review, testing and continuous delivery.

Naval Surface Warfare Center Dahlgren | Computer Scientist — Jun 2021 – May 2022

- Built Python data-labeling and processing pipelines supporting TensorFlow and YOLOv5 model training with engineering and research teams.

Northrop Grumman and Peraton | Software Engineer — May 2018 – Jun 2021

- Developed Python ETL tools for a research project. On a separate project, maintained C++ simulation software, investigated defects and performed integration testing.
- Continued employment through the transfer of the Northrop Grumman business to Peraton; supported builds, version control and user training.

SELECTED AI PROJECTS

LLM Tool Calling Lab | Python, Ollama, scikit-learn

- Built an experimental local chatbot to explore how an LLM can use specialized ML models as tools. It answers questions about numerical data, compares methods and shows recorded results, with limits on model runs and tool calls.
  github.com/lpaynedev/llm-tool-calling-lab

Earth Science Metadata Agent | Python, FastAPI, LangGraph

- Built an independent prototype for finding Earth science datasets from plain-language questions. It searches NASA's public metadata API, retrieves related context with Chroma and summarizes returned records using FastAPI and LangGraph.
  github.com/lpaynedev/NASA_CMR_AGENT

ADDITIONAL EXPERIENCE AND EDUCATION

Freelance AI Software Development

Computer science tutoring in Python, algorithms, object-oriented programming and debugging.

University of Mary Washington | B.S. Computer Science; B.A. Music, 2018